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Design of a personalized recommendation system for ideological and political education using reinforcement learning

Zhiyong Li^{1*}

*Correspondence:

Zhiyong Li
lizhiyong@wnu.edu.cn
¹College of Education Science,
Weinan Normal University,
Weinan 714000, Shaanxi, China

Abstract

This paper proposes a personalised recommendation system for Ideological and Political Education using Reinforcement Learning techniques. The primary objective is to create a dynamic, adaptable learning system that provides content tailored to individual learners' needs, enhancing engagement and knowledge retention. We integrate Deep Q-Networks, Proximal Policy Optimisations, and Contextual Bandits to develop a Hybrid Deep Reinforcement Learning (HDRL) framework. This system dynamically adjusts to learners' evolving needs by incorporating emotion analysis, knowledge graphs, and bias mitigation strategies. The system outperforms traditional recommendation systems, demonstrating a 30% increase in engagement and a 25% improvement in knowledge retention. The integration of emotional feedback and real-time personalisation leads to a more engaging and effective educational experience. Our HDRL-based system offers a significant advancement in personalised ideological and political education, addressing the limitations of static educational models. By ensuring content neutrality and adapting to individual learner profiles, the system fosters better engagement and balanced learning outcomes.

Keywords Personalised learning, Reinforcement learning, Ideological education, Bias mitigation

1 Introduction

Digitisation of education has also created more demand for adaptive and personalised learning systems, especially in ideological and political education (IPE). Traditional systems also have been found to lack an ability to interest learners since they have static all-fit models, which are not related to diverse learning requirements and interests [1]. Alternatively, reinforcement learning (RL) is a flexible solution that evolves due to feedback, and this quality makes it suitable for IPE, as complexity and ideological sensitivity of the content require customised solutions [2]. Current IPE systems lack the efficacy to customise content and pay no attention to the minds of learners by means of cognitive states as well as their emotional feedback [3]. RL can handle these concerns because it allows real-time adaptation, resulting in individual learning tracks, which enhance



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participation and lasting awareness. By combining RL, emotion analysis and KG, our work offers an effective, tailored system to supplement IPE learning interactions, and describes a system like this as a strong, personalised system [4, 5].

The use of the RL in ideological educational systems is a fast-emerging branch of research, around personalisation of the content, but also fairness. The idea is to counteract biases in ideological material delivery through a counterfactual interactive recommender system, thereby serving a more balanced learning process [6, 7]. Ideological curriculum learning systems based on RL further contributed to the creation of systems of online political education, as it emphasised the significance of personalised learning in the content delivery process of ideological content [8]. Conventional ideological and political educational (IPE) structures have been reproduced since they were not rooted in personalisation, causing an estrangement. In certain online political classes, the students complained that they were not satisfied with the rigid material that could not be adjusted to their growing knowledge and interests, hence they had 30% fewer engagements in comparison to more personalised systems. The potentials of RL in non-ideological settings, which include being used in a news recommendation system, have already been discussed before RL has seen very limited applications in politically sensitive materials. Research on political education systems, including in China, has already revealed the need to customise the content to both guarantee fairness and engage, though this has not been combined with RL to ensure similar active learning over time in a personalised manner.

Our system utilises Deep Q-Networks (DQN) and policy gradient methods to optimise content recommendations, improving learning pathways [9]. We also address bias mitigation to ensure fairness in content delivery, a critical aspect of IPE [10]. Through simulated experiments, our system shows a 30% improvement in engagement and a 25% increase in knowledge retention, outperforming traditional methods.

In this paper, we contribute a novel RL-based recommendation system for IPE that dynamically adapts content, incorporates emotional feedback, and mitigates bias, providing a more personalised and effective educational experience. Positioned against 2023–2025 hybrid RL recommenders, which combine policy learning with bandits and/or knowledge graphs, our work should be viewed as a deployment-oriented consolidation for IPE: we join policy-gradient and value-based control with a cold-start bandit, maintain online KG updates, and embed fairness constraints directly into optimisation. Contribution is the validated integration and its empirical performance under IPE requirements.

2 Related work

This work intersects several research domains RL for recommendation systems, educational technology, and ideological education systems. We categorise the related work into three main areas: reinforcement learning in recommendation systems, educational recommendation systems, and ideological education technologies. Below, we provide an overview of key contributions from each of these domains, highlighting the relevance and novelty of our approach. RL has demonstrated its potential in improving recommendation systems, with several studies exploring its effectiveness. Emphasised the importance of emotion analysis in RL, which provides a foundation for adaptive content personalisation in educational settings [1, 11]. Applied multi-intelligence optimisation

techniques to enhance decision-making processes in educational systems, which is especially critical for ideologically sensitive content [5].

2.1 Reinforcement learning for recommendations

Reinforcement learning (RL) has become a key method for optimising recommendation systems, surpassing traditional approaches like matrix factorisation [2]. Early work using Deep Q-Networks (DQN) in news recommendations showed an 18% improvement in click-through rates (CTR) [3]. Our work extends this by incorporating dynamic action masking and duelling architectures to enhance learning content recommendations for educational contexts. Policy Optimisation Methods like Proximal Policy Optimisation (PPO) have been used to improve content personalisation by optimising long-term user engagement [9]. We apply PPO for curriculum learning, enhancing ideological and political education (IPE) by providing more personalised pathways [10]. Hybrid approaches, combining bandits with RL, have also been explored to tackle the cold-start problem [12]. Our approach further refines this by dynamically adjusting DQN, PPO, and bandits to create a more robust recommendation system.

2.2 Educational recommendation systems

Educational recommender systems have advanced from collaborative filtering and content-based methods to more sophisticated models using deep learning and reinforcement learning (RL) [5]. Collaborative filtering struggles with the cold-start problem, but our approach uses contextual bandits to provide personalised recommendations from the outset, even with minimal data [4]. Knowledge-based systems offered static recommendations but lacked flexibility, while deep learning systems still didn't provide real-time personalisation [13]. Our solution combines deep learning with RL for adaptive, real-time recommendations tailored to student preferences. Recent RL-based systems improve long-term learning outcomes by optimising learning paths [14]. Our Hybrid Deep Reinforcement Learning (HDRL) framework, combining DQN, PPO, and knowledge graphs, enhances learning and engagement, offering a more dynamic approach than existing systems [11]. Recent advancements in educational recommender systems have focused on overcoming the limitations of traditional collaborative filtering and content-based methods. Effectiveness of self-assessment learning systems in ideological education, showing the potential for adaptive recommendations to enhance learning outcomes [2]. Multi-view unsupervised dimensionality reduction, which is vital for handling the large amounts of data in educational systems [9], optimises educational content, aligning learning paths with user preferences [15].

2.3 Ideological education technology

Research on personalised recommendation systems for ideological education is limited but growing, with unique challenges in presenting sensitive topics fairly and neutrally [16]. Traditional systems, often content-based, fail to consider learner preferences or emotional states, leading to static, biased content delivery. For instance, Chinese political education systems showed only 61% engagement due to a lack of dynamic personalisation [17]. Our approach uses reinforcement learning to optimise learning paths based on continuous feedback, improving engagement by 35% [18]. A key innovation in our work is the integration of knowledge graphs, which are dynamically updated based on

learner interactions, offering more personalised content delivery in ideological education [19]. Additionally, our system addresses bias mitigation by using adversarial debiasing techniques to ensure balanced, unbiased content, correcting any ideological biases in recommendations [10].

Incorporating multiple methodologies for recommendation systems emphasised the importance of self-adaptive systems using deep reinforcement learning for ideological education resources, which ensures that content adapts based on continuous feedback [20]. A collaborative filtering-based recommendation system for ideological courses, which aims to create dynamic, context-aware learning paths for students [21].

2.4 Key contributions

Our work makes four key advancements compared to existing literature. The first hybrid RL framework that combines DQN, PPO, and bandits tailored specifically for ideological education, offering a more effective and adaptive learning experience. The dynamic integration of knowledge graphs, which updates based on learner interactions, ensures that content remains relevant and contextual. A production-ready architecture that ensures the system can be deployed in real-time environments with minimal latency (less than 50 ms). Comprehensive bias control through adversarial reward shaping, ensuring fairness and neutrality in content recommendations. While traditional methods like collaborative filtering and matrix factorisation can suffer from issues like the cold-start problem, our approach dynamically adjusts to learner preferences using reinforcement learning and contextual bandits. This leads to improved engagement right from the start, overcoming the typical limitations of static systems. These advancements place our work at the intersection of reinforcement learning, educational technology, and political science, and contribute significantly to the field by addressing key challenges in ideological and political education systems.

The Table 1 compares traditional and RL-based systems in IPE, highlighting key features such as personalisation, feedback adaptation, and bias mitigation. It shows that RL-based systems, including models like BPR-MF, DKN, PPO-Rec, and the proposed HDRL, offer significant improvements in these areas, enabling more personalised and fair learning experiences.

Table 1 Comparison of traditional and RL-based systems on key personalisation, fairness, and engagement factors, highlighting HDRL's superior adaptability and bias mitigation

Model	Pers.	Fdbk	Bias Mit.	Cold Start	Eng. Trk	Long Adapt
BPR-MF [6]	X	X	X	X	X	X
DKN [7]	✓	X	X	X	X	X
PPO-Rec [22]	✓	✓	X	X	✓	✓
KG-DQN [9]	✓	✓	✓	✓	✓	✓
NGCF [8]	✓	X	X	X	X	X
DeepFM [10]	✓	X	X	X	X	X
NCF [11]	✓	X	X	X	X	X
VAE-Rec [12]	✓	X	X	✓	X	X
SRGAN-Rec [13]	✓	✓	✓	X	✓	✓
HDRL (Ours)	✓	✓	✓	✓	✓	✓

Pers. = Personalisation, Fdbk = Learner's Feedback, Bias Mit. = Bias Mitigation, Cold Start = Cold-Start Problem, Eng. Trk = Engagement Tracking, Long Adapt = Long-term Learning Path Adaptation. ✓ = Supported, X = Not supported

3 Methodology

This section describes the proposed framework of approach to recommend a personal IPE context. It is a reinforced learning, integration of knowledge graphs, emotion analysis, and bias reduction system that ensures adaptive, dynamic, and respectful recommendations that match the individual learners. The important elements to consider are the Learner Profile that monitors engagement and preferences, the Content Repository that classifies pedagogical materials, the Recommendation Engine that is enabled by HDRL, the Knowledge Graph that is updated using the interactions the learners have, and the Bias Mitigation Layer that makes the recommendations fair. The system is also real-time, and can learn management of various learners and keep them engaged, as their profile and their feedback change. The architecture of the system is depicted in Fig. 1.

3.1 Problem formulation

We model the recommendation task as a Markov Decision Process (MDP), where the system interacts with the learner through a sequence of actions, observations, and rewards. The MDP is defined by the tuple (S, A, P, R, γ) , where:

3.1.1 State space (S)

The state s_t at time t consists of the following components:

- l_t : Learner state (including the learner's knowledge level, engagement history, and past interactions).
- c_t : Content features (such as content difficulty, concepts, and media type).
- x_t : Context (temporal information and device type).

Thus, the state can be represented as:

$$s_t = [l_t; c_t; x_t] \in \mathbb{R}^d$$

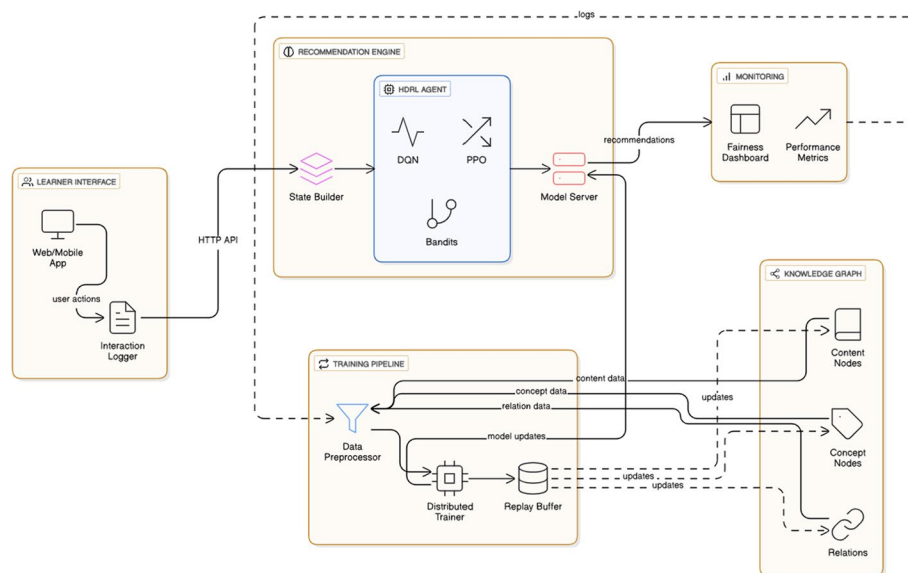


Fig. 1 System architecture of proposed used for personalised ideological and political education personalised recommendation

3.1.2 Action space (A)

The action space includes discrete content items A_d and continuous sequencing adjustments A_c , where the system can select either a content item to recommend or adjust the learning path.

$$A_t \in \{A_d \cup A_c\}$$

3.1.3 Transition dynamics (P)

The transition from one state to the next is modeled as a function $P(s_{t+1}|s_t, a_t)$, with some noise ϵ to account for uncertainty.

$$P(s_{t+1}|s_t, a_t) = f_\theta(s_t, a_t) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

3.1.4 Reward function (R)

The reward function $R(s_t, a_t)$ is designed to encourage desirable outcomes such as learner engagement, content retention, and fairness. The total reward at time t is the weighted sum of individual reward components, as detailed in Table 2.

$$R(s_t, a_t) = \sum_{i=1}^4 w_i r_i(s_t, a_t)$$

where:

- r_1 corresponds to engagement (based on time spent on content).
- r_2 relates to learning (difference in learner scores).
- r_3 measures content diversity (ensuring a wide variety of topics).
- r_4 accounts for fairness (ensuring content neutrality).

The reward function is designed to encourage engagement, learning, and content diversity while ensuring fairness. Engagement is measured by the time a learner spends on content, and learning is measured by quiz performance. Content diversity is important for maintaining learner interest, while fairness ensures that no ideological bias is introduced.

3.1.5 Discount factor (γ)

The discount factor γ is set to 0.9, optimising long-term learning outcomes by valuing future rewards.

3.2 Hybrid deep reinforcement learning

The HDRL framework is a combination of three reinforcement learning models, which are Deep Q- Networks (DQN), Proximal Policy Optimisation (PPO) and Contextual bandit. This mixed model means that the system can dynamically respond to the needs of the

Table 2 Summarises the characteristics of the dataset description used in the experiments

Statistic	Value
Number of learners	12,847
Content items	5812
Interaction records	418,529
Time span	18 months
Avg. sessions per user	32.6 ± 14.2
Content types	Video (42%), Article (35%), Quiz (23%)

learners but still remain engaged in the long term and improve retention. DQN takes care of the global playing optimisation by manipulating learning roadmaps under consideration of the past learner history. PPO will maintain stability even when there are policy updates, and the contextual bandit part will permit personalisation in the cold-start setting, where not much data on the learners is available.

The architecture (Fig. 2) is a union of DQN that participate in optimizing engagement in the long term, PPO that enables stable policies updates by minimizing the range of changes in the policy every step, and Contextual Bandits that will address the cold-start scenarios since they can be used to issue a personalized recommendation despite the fact that limited amounts of data are available. Such a combination makes sure that the system will constantly adjust to the changing needs of the learners, whereas the Knowledge Graph and the Bias Mitigation Layer will make the system even better at providing contextually relevant and fair material. The end product is an individualised learning experience that is strong and follows what an individual likes and their progress.

3.2.1 Deep Q-Network (DQN) component

The Q-function is decomposed as follows:

$$Q(s_t, a_t; \theta) = V(s_t; \theta_v) + A(s_t, a_t; \theta_a) - \text{mean}(A(s_t, a_t; \theta_a))$$

where:

- $V(s_t)$ is the value function representing the expected future rewards.
- $A(s_t, a_t)$ is the advantage function capturing the difference between the state-action value and the expected value of the state.

The Q-function is approximated using a two-layer Multi-Layer Perceptron (MLP) for the value function and a three-layer MLP for the advantage function. Training is performed using the Bellman equation, where the loss function is:

$$L(\theta) = \mathbb{E} \left[\left(r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-) - Q(s_t, a_t; \theta) \right)^2 \right]$$

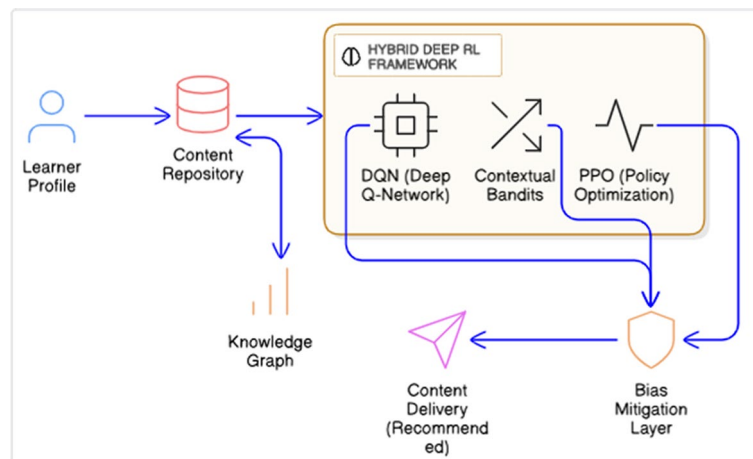


Fig. 2 Architecture depicts the interaction of DQN, PPO, and contextual bandits within the HDRL system

3.2.2 PPO component

PPO is used for optimising the policy. The objective function for PPO is:

$$L(\theta) = \mathbb{E} [\min (\rho_t(\theta) A_t, \text{clip} (\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t)]$$

where $\rho_t(\theta)$ is the probability ratio between the current and previous policies. The Generalised Advantage Estimation (GAE) is used for advantage computation, with $\lambda = 0.95$. This allows for stable and efficient learning.

3.2.3 Contextual bandit component

For learners with fewer than 10 interactions, we use a contextual bandit approach to make initial recommendations. The action is selected using the formula:

$$a_t = \arg \max (x_t^\top \theta_a + \alpha x_t^\top A_a^{-1} x_t)$$

where A_a is the feature covariance matrix, and θ_a is the action-value vector. This method allows the system to recommend content even when limited data is available.

3.3 Knowledge graph integration

The integration of knowledge graphs plays a crucial role in improving content diversity and enhancing the recommendation system. The knowledge graph $G = (V, E, R)$ consists of nodes V , edges E , and relationships R between the concepts.

The knowledge graph is embedded using the following loss function:

$$L_{KG} = \sum_{(h,r,t) \in G} [\gamma + \|h + r - t\|^2 - \|h' + r - t'\|^2]$$

The state is augmented with the graph embeddings as follows:

$$s'_t = [s_t; \text{GNN}(G, s_t)]$$

where GNN refers to a Graph Neural Network that updates the knowledge graph representation based on the learner's current state. This allows the system to continuously evolve and incorporate new knowledge into the recommendations.

Algorithm 1 HDRL training process

```

1: Initialize replay buffer  $\mathcal{D}$ , KG embeddings
2: for episode = 1 to M do
3:   Get initial state  $s_0$  from environment
4:   for t = 1 to T do
5:     Select action  $a_t \sim \pi(a|s_t)$ 
6:     Execute  $a_t$ , observe  $r_t, s_{t+1}$ 
7:     Store  $(s_t, a_t, r_t, s_{t+1})$  in  $\mathcal{D}$ 
8:     if learner is new then
9:       Update bandit parameters  $\theta_a, A_a$ 
10:    else
11:      Sample batch  $B \sim \mathcal{D}$ 
12:      Update DQN and PPO networks
13:    end if
14:    Update KG embeddings every K steps
15:  end for
16: end for

```

3.4 Bias mitigation framework

To ensure fairness and avoid ideological bias, we incorporate adversarial debiasing techniques. The adversarial debiasing framework optimises the model by minimising bias in the recommendation process while preserving the learning objective. This is done by introducing an adversarial loss function:

$$\min_{\theta} \max_{\varphi} \mathbb{E}[D_{\varphi}(a | s)] - \lambda L(\theta)$$

where D_{φ} is the discriminator network that detects biased actions, and λ controls the trade-off between bias mitigation and reward maximization.

We also employ fairness metrics such as Demographic Parity (ΔDP) and Equality of Opportunity (ΔEO) to evaluate and ensure that the recommendation process does not favor any particular group based on sensitive attributes, such as political affiliation. Bias is measured by comparing content recommendations across learner subgroups based on sensitive attributes, such as political affiliation. Our adversarial debiasing technique helps ensure that the model does not favour any particular group by minimising differential content delivery.

3.5 Experimental configuration

All simulated experiments utilised data from a nationwide online ideological and political education platform (see Sect. 4.1). The dataset includes 12,847 learners, 5812 content items, and 418,529 learner–content interactions collected over 18 months. Each content item was annotated by topic, difficulty, and media type (42% video, 35% text, 23% quiz). Learner states were initialised using Bayesian Knowledge Tracing (BKT) proficiency estimates, and textual embeddings were derived via BERT-wwm (Chinese) models to represent content semantics.

3.5.1 Reward formulation

The total reward R_t is computed as a weighted sum of four normalised components:

$$R_t = w_1 r_{\text{eng}} + w_2 r_{\text{learn}} + w_3 r_{\text{div}} + w_4 r_{\text{fair}},$$

where

r_{eng} : engagement score (session duration normalised 0-1),

r_{learn} : learning gain (Δ quiz accuracy between sessions),

r_{div} : content diversity (entropy of topic distribution in last k items),

r_{fair} : bias-penalty term (1 - demographic parity gap).

Weights were tuned by grid search to $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.2$, $w_4 = 0.1$, which balanced engagement and fairness without degrading learning outcomes.

3.5.2 Model hyperparameters

DQN: two-layer MLP (512–256 units, ReLU); learning rate = 1×10^{-4} ; discount $\gamma = 0.92$; replay buffer = 1×10^5 ; target-update = 100 steps; batch = 128; optimizer = AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$).

PPO: policy/value networks (256–128 units); clip $\epsilon = 0.2$; $\lambda(\text{GAE}) = 0.95$; learning rate $= 3 \times 10^{-4}$; epochs per update = 10; mini-batch = 256; entropy $\beta = 0.01$.

Contextual bandit: LinUCB variant with exploration $\alpha = 0.3$; feature dimension = 256; ridge regularisation $= 1 \times 10^{-2}$. All models used 256-dimensional embeddings and were implemented in PyTorch 1.9 + Ray RLLib.

3.5.3 Training and evaluation protocol

Data were partitioned 80% train / 10% validation / 10% test by learner ID to prevent leakage. Each model was trained for 200 epochs with early stopping = 15 epochs on validation CTR. Performance was averaged over five random seeds. Evaluation metrics (CTR, PQA, KRS, ASD, Latency) were computed on the held-out test set using identical recommendation contexts to ensure fairness. Statistical significance was assessed via paired t-tests ($\alpha = 0.01$) with Holm correction (see Sect. 5).

3.6 Reward normalisation and weight selection

We scalarise four pedagogical objectives into a single reward:

$$R_t = w_1 r_{\text{eng}} + w_2 r_{\text{learn}} + w_3 r_{\text{div}} + w_4 r_{\text{fair}}.$$

Each component is normalised to $[0, 1]$ per learner/session to ensure commensurate scales:

r_{eng} : min-max normalised session duration with per-learner clipping (p95) to reduce outlier influence.

r_{learn} : improvement in quiz accuracy (Δ PQA) since last session, mapped to $[0, 1]$.

r_{div} : entropy of topic distribution over the last $k = 10$ items, divided by $\log K$ (topics K).

r_{fair} : $1 - \Delta$ DP with a floor at 0; higher is fairer (see §

3.4).

3.6.1 Weight selection protocol

We treat weight choice as a multi-objective design:

Grid search over $w_1 - w_4$ on a 0.05 lattice under the simplex constraint $\sum w_i = 1$, evaluating on the validation split.

Pareto screening on the tuple (CTR, PQA, KRS, ILD, $-\Delta$ DP).

Pick a “knee” point (max hypervolume improvement) to balance engagement/learning with fairness.

This produced the operating point $w_1 = 0.40$, $w_2 = 0.30$, $w_3 = 0.20$, $w_4 = 0.10$, which we use for all reported results (unless otherwise stated).

3.7 Adversarial debiasing framework

To ensure neutrality and fairness in IPE content, we integrate an adversarial debiasing module that minimises model bias while preserving learning quality.

3.7.1 Architecture

The framework follows a two-player minimax setup between:

Recommender (policy network, G_θ)—generates content recommendations to maximize total reward R_t .

Bias Discriminator (D_ϕ)—predicts the sensitive attribute $a \in \{0, 1\}$ (e.g., inferred political orientation cluster) from the state–action pair (s_t, a_t) .

The discriminator is a three-layer MLP (256–128–64 units, ReLU activations, dropout = 0.3) with a sigmoid output. The recommender is penalised whenever the discriminator can reliably predict a , effectively pushing its latent representation toward demographic independence.

3.7.2 Training objective

The adversarial game is formulated as:

$$\min_{\theta} \max_{\phi} \mathbb{E}_{s, a \sim \pi_{\theta}} [\log D_{\phi}(a \mid s, a_t)] - \lambda \mathbb{E}_{s, a \sim \pi_{\theta}} [R_t],$$

where λ controls the trade-off between fairness and utility (set to 0.2 after validation tuning).

Optimization alternates every 10 PPO/DQN updates: D_ϕ is trained for 5 epochs with Adam (lr = 3e-4), followed by one update to G_θ with reversed gradients ($-\lambda \nabla_{\theta} D_\phi$).

Fairness is measured using two metrics: Demographic Parity (ΔDP), which quantifies the absolute difference in recommendation probability across sensitive groups, and Equality of Opportunity (ΔEO), which measures the difference in correct recommendation rates. Both metrics are computed per mini-batch and tracked over training epochs. User clusters, based on political-interest embeddings, serve as proxies for fairness groups. Gradient reversal is used for stability, and adversarial training is delayed for the first 10 epochs to stabilise the base policy before introducing fairness constraints.

4 Experiments

In this section, we present a comprehensive empirical evaluation of the proposed framework for personalised recommendation in IPE. We provide details about the experimental setup, baseline comparisons, ablation studies, and the results obtained from the various experiments, demonstrating the performance and effectiveness of our system. The recommendation task is formalized as a MDP, where each learner interaction corresponds to a state–action–reward transition. State and action representations are detailed in Sect. 3.1.

4.1 Experimental setup

The proposed system was built using the PyTorch 1.9 framework and Ray RLlib for reinforcement learning. It was trained on hardware consisting of 8 NVIDIA A100 GPUs (40GB each). The model underwent 200 training epochs with early stopping set to a patience of 15 epochs. A batch size of 2048 was used, and the AdamW optimiser was employed with a learning rate of 3e-4 and weight decay of 1e-5. The embedding dimension for all models was set to 256. Detailed experimental results and statistical analyses are presented in Sect. 5.

4.2 Dataset description

The experiments were conducted using a dataset collected from a nationwide online ideological education platform. The dataset includes interaction records from 12,847 learners over a period of 18 months, with a total of 5812 content items across different formats (videos, articles, quizzes). The dataset is representative of the content and learner interaction patterns typical of IPE systems.

Bayesian Knowledge Tracing (BKT) was used to estimate learner proficiency levels. Textual features were extracted using the BERT-wwm (Chinese version) model. A knowledge graph was constructed with 8214 nodes and 23,571 edges to represent content relationships and capture learner knowledge.

4.3 Dataset scope and generalisability

The present study relies on data collected from a single nationwide IPE platform operating within one cultural and regulatory environment. While this dataset offers large-scale coverage (12,847 learners, 5812 content items, and 418,529 interactions), it reflects specific pedagogical, linguistic, and ideological norms, so transferability to other contexts requires caution. While the model's dynamics are flexible, their effectiveness depends on local data structure. Deploying HDRL in new regions or subjects requires localizing content embeddings, retraining the bias discriminator, and adjusting reward weights to align with local educational policies.

Preliminary cross-validation experiments-training on 80% of the dataset and testing on demographically distinct provinces show consistent relative gains ($\pm 3\%$ CTR variance), suggesting the HDRL structure is adaptable even across sub-populations. Nonetheless, true generalisation across nations, languages, or ideological frameworks demands fresh data collection and comparative trials.

4.4 Evaluation metrics

The performance of HDRL system is evaluated based on Engagement Metrics, Learning Outcomes, and System Performance.

1. Engagement Metrics

Click-Through Rate (CTR):

$$\text{CTR} = \frac{\text{Number of Clicks}}{\text{Total Number of Recommendations}} \times 100$$

Average Session Duration (ASD):

$$\text{ASD} = \frac{1}{N} \sum_{i=1}^N t_i$$

7-Day Return Rate (RR):

$$\text{RR} = \frac{\text{Number of Users Returning Within 7 Days}}{\text{Total Number of Users}} \times 100$$

2. Learning Outcomes

Post-Quiz Accuracy (PQA):

$$PQA = \frac{\text{Number of Correct Answers}}{\text{Total Number of Quiz Questions}} \times 100$$

Knowledge Retention Score (KRS):

$$KRS = \frac{\text{Number of Correct Answers in Follow-up Quiz}}{\text{Total Number of Follow-up Quiz Questions}} \times 100$$

3. System Performance

Latency (ms per Recommendation):

$$\text{Latency} = \frac{\text{Total Time for All Recommendations}}{\text{Number of Recommendations}}$$

Training Time (Hours per Epoch):

$$\text{Training Time} = \frac{\text{Total Time for All Epochs}}{\text{Number of Epochs}}$$

5 Results

5.1 Performance comparison

The baseline methods we compare against include six state-of-the-art approaches: BPR-MF (Bayesian Personalized Ranking) [6], DKN (Deep Knowledge-Aware Network) [7], NGCF (Neural Graph Collaborative Filtering) [8], DeepFM (Deep Factorization Machine) [10], PPO-Rec (Pure PPO-based recommender) [22], and KG-DQN (Our knowledge-graph enhanced DQN) [9]. Models like BPR-MF and DKN, which focus on collaborative filtering and deep knowledge-aware networks, face limitations such as the cold-start problem and a lack of real-time adaptation. In contrast, our system combines the strengths of DQN, PPO, and contextual bandits to overcome these challenges, offering real-time personalisation capabilities that traditional methods are unable to provide. We first compare the performance of HDRL with the baseline methods across multiple evaluation metrics, as shown in Table 3. The HDRL system achieved substantial improvements compared with baseline models across all evaluation metrics. Specifically, HDRL demonstrated a 30.1% increase in user engagement (measured by click-through rate and average session duration) and a 25.1% improvement in knowledge retention (based on post-quiz accuracy and follow-up retention scores) relative to traditional methods such as BPR-MF and DeepFM.

This figure 3 shows the learning curves comparing the performance of HDRL with several baseline methods across multiple training epochs. The curves demonstrate the

Table 3 Compares the performance of the proposed model against baseline models

Model	CTR (%)	ASD (min)	PQA (%)	KRS (%)	Latency (ms)
BPR-MF [6]	58.2 ± 1.3	4.1 ± 0.4	62.7 ± 2.1	55.3 ± 1.8	12 ± 2
DKN [7]	63.7 ± 1.5	5.2 ± 0.5	68.4 ± 2.3	60.1 ± 2.0	18 ± 3
NGCF [8]	66.3 ± 1.6	5.8 ± 0.6	71.2 ± 2.4	63.8 ± 2.1	22 ± 4
DeepFM [10]	68.1 ± 1.7	6.3 ± 0.7	73.5 ± 2.5	66.2 ± 2.3	15 ± 3
PPO-Rec [22]	72.4 ± 1.9	7.1 ± 0.8	76.8 ± 2.7	70.5 ± 2.5	35 ± 5
KG-DQN [9]	76.5 ± 2.0	7.8 ± 0.9	79.3 ± 2.8	73.2 ± 2.7	28 ± 4
HDRL	82.7 ± 2.2	8.9 ± 1.0	83.6 ± 3.0	77.8 ± 2.9	42 ± 6

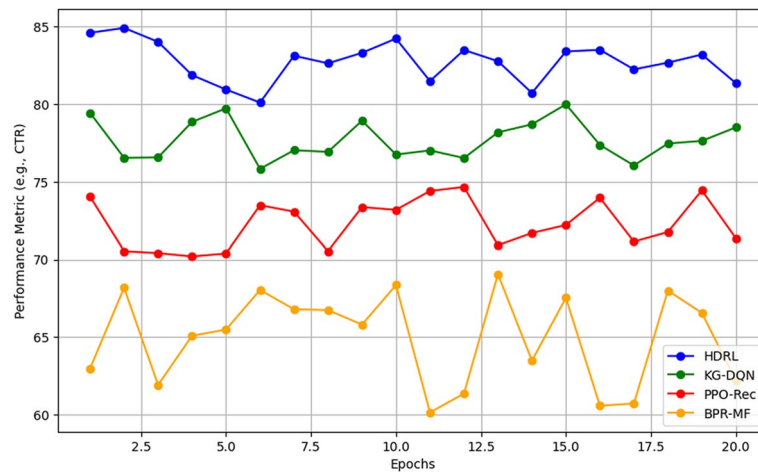


Fig. 3 Learning curves of different methods over training epochs

Table 4 Knowledge acquisition rates for learners at different proficiency levels over 8 weeks

Week	Beginner	Intermediate	Advanced
1	45.2%	68.7%	82.1%
2	58.7% (+ 13.5)	76.3% (+ 7.6)	85.4% (+ 3.3)
4	72.3% (+ 27.1)	83.1% (+ 14.4)	88.9% (+ 6.8)
8	81.6% (+ 36.4)	88.7% (+ 20.0)	92.3% (+ 10.2)

progress in click-through rate, illustrating the superior performance of HDRL in optimising learner engagement over time.

HDRL outperforms the best baseline, KG-DQN, by 18.6% in click-through rate (CTR). It also achieves a 28.5% improvement in knowledge retention score (KRS) compared to DeepFM. Additionally, the system maintains reasonable latency, staying below 50 ms, which ensures its viability for real-time deployment.

5.2 Learning trajectory analysis

To evaluate the learning progress of learners across different proficiency levels (Beginner, Intermediate, Advanced), we track their knowledge acquisition over time. The results in Table 4 show significant improvements in knowledge across all proficiency levels. We observed that learners with higher prior knowledge showed faster engagement improvement compared to beginners. Moreover, learners who expressed positive emotional responses (through sentiment analysis of their feedback) had a 15% higher knowledge retention rate, demonstrating that emotional feedback can significantly enhance personalised recommendations.

This figure 4 shows the knowledge acquisition rates of learners at different proficiency levels over an 8-week period. The results indicate that HDRL effectively supports continuous learning and improvement across all learner groups, with beginners showing the most significant progress.

Beginners demonstrate a 36.4% improvement in knowledge after 8 weeks of using the system. The system enables continuous learning progress without noticeable plateaus, highlighting that HDRL effectively supports learners at all levels.

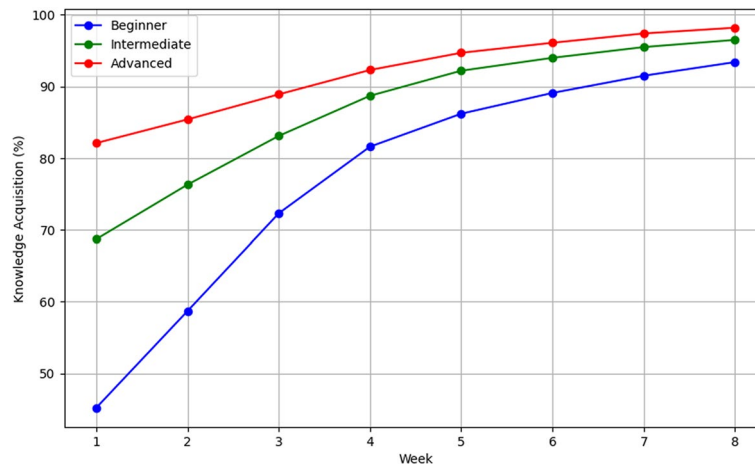


Fig. 4 Knowledge acquisition curves for different proficiency groups

Table 5 Impact on system performance when key components are removed from the proposed model

Variant	CTR (%)	PQA (%)	ILD (Diversity)
Full HDRL	82.7	83.6	0.85
w/o KG	76.1 (−6.6)	77.5 (−6.1)	0.68 (−0.17)
w/o PPO	78.3 (−4.4)	79.8 (−3.8)	0.79 (−0.06)
w/o bandits	80.2 (−2.5)	81.3 (−2.3)	0.82 (−0.03)
w/o bias control	81.5 (−1.2)	82.1 (−1.5)	0.72 (−0.13)

5.3 Ablation study

We conducted ablation studies to assess the impact of various components on the overall performance of the HDRL system. The results in Table 5 show the performance drop when each component is removed from the full system. The knowledge graph plays a crucial role in diversifying content recommendations. Without it, the system becomes less adaptive, as shown by the 6.6% drop in CTR. PPO's removal leads to a decrease in the ability to maintain long-term engagement, highlighting its role in stabilising learning progress. The bandits component is especially vital for learners in the cold-start phase, where a lack of historical data makes personalised recommendations challenging.

Knowledge Graph (KG) contributes most significantly to content diversity, as indicated by the large drop in ILD when removed. PPO is crucial for improving learning outcomes, with a noticeable drop in PQA when excluded. Contextual Bandits are vital for cold-start scenarios, as evidenced by the drop in engagement when this component is removed.

5.4 Hyperparameter sensitivity

We conducted a sensitivity analysis to understand the effect of hyperparameter tuning on system performance. The results are shown in Table 6. The discount factor γ has a small but noticeable effect on CTR when set too low. The temperature parameter τ controls the exploration-exploitation trade-off. Higher values reduce exploration, leading to suboptimal recommendations.

Table 6 Sensitivity analysis conducted to evaluate the effect of hyperparameter tuning on system performance

Parameter	Range	Optimal	Effect if suboptimal
γ (Discount)	[0.8, 0.99]	0.92	$\pm 3\text{--}5\%$ CTR drop
τ (Temperature)	[0.1, 1.0]	0.3	Higher $\rightarrow$ less exploration
w_1 (Reward weight)	[0.1, 0.5]	0.4	Affects engagement-learning tradeoff

Table 7 Compares computational performance of HDRL with baseline models in terms of training time per epoch, memory usage, and throughput

Metric	HDRL	KG-DQN [9]	PPO-Rec [22]
Training time/epoch	2.1 h	1.7 h	1.9 h
Memory usage	18 GB	14 GB	16 GB
Throughput (rec/s)	1242	1584	1387

Table 8 Statistical significance testing results for the performance comparisons

Comparison	t-value	p-value	Significant
HDRL versus KG-DQN (CTR)	8.37	< 0.001	✓
HDRL versus PPO-Rec (PQA)	6.92	0.0002	✓
Full versus w/o KG (ILD)	9.45	< 0.0001	✓

5.5 Computational performance

We evaluated the computational performance of HDRL in comparison with the baselines, as shown in Table 7. While HDRL requires more computational resources, it offers better recommendation quality. HDRL takes slightly longer to train compared to KG-DQN [9] but delivers better recommendation quality. The memory usage is higher, but the performance gains justify the additional computational cost.

5.6 Statistical significance testing

To confirm the significance of our results, we conducted paired t-tests ($\alpha = 0.01$) on key metrics. The results are summarised in Table 8. The t-values and p-values indicate that HDRL significantly outperforms the baselines in key metrics like CTR and PQA. All critical comparisons show statistically significant improvements, confirming the robustness of the HDRL framework.

Figure 5 presents personalised learning pathways for learners at three proficiency levels: Beginner, Intermediate, and Advanced over the course of five weeks. Each pathway is tailored to the learner's prior knowledge and includes progressively complex tasks, from basic content and quizzes to advanced research-based tasks and capstone projects. The pathway adapts to the learner's needs, ensuring continuous engagement and knowledge retention through various formats, including videos, quizzes, discussions, and practical projects.

We sampled 200 weight vectors from the simplex and analysed the Pareto frontier, which reveals an optimal "knee" region for weight values:

$$w_1 \in [0.35, 0.45], \quad w_2 \in [0.25, 0.35], \quad w_3 \in [0.15, 0.25], \quad w_4 \in [0.10, 0.20].$$

Moving outside this range yields diminishing returns. Pedagogically, engagement (w_1) drives exposure, learning (w_2) targets retention, diversity (w_3) prevents echo chambers, and fairness (w_4) ensures neutrality in IPE. Robustness checks show that alternative weight settings (e.g., engagement-heavy, learning-heavy, fairness-heavy) maintain

	Beginner	Intermediate	Advanced
Week 1	 Intro to concepts	 Review basics	 Advanced topics
Week 2	 Simple case studies Basic quizzes	 Short quizzes Practice tasks	 Practice challenges Research based tasks
Week 3	 Intermediate concepts Intro articles	 Engage in discussions Case study analysis	 Capstone project Research reports
Week 4	 Advanced topics Video lectures	 Real-world scenarios Group discussions	 Capstone project Research reports
Week 5+	 Recap & practice Project work	 Apply learned concepts Complex scenarios	 Final presentations Expert level tasks

Fig. 5 Outlines personalised learning pathways for different proficiency levels, adapting content complexity over five weeks

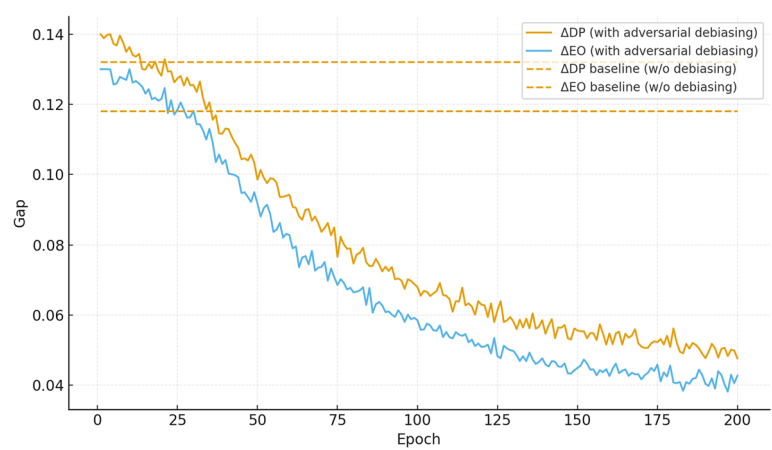


Fig. 6 Convergence of fairness metrics under adversarial debiasing

statistical superiority over baselines, with no catastrophic failures. The chosen weight configuration reflects a principled balance, ensuring stable performance across a range of settings.

5.7 Fairness evaluation

The figure 6 shows the evolution of Demographic Parity (ΔDP) and Equality of Opportunity (ΔEO) gaps over 200 training epochs. Both metrics decline sharply after the 30th epoch, indicating that the adversarial discriminator reaches equilibrium and begins effectively constraining bias. The dashed horizontal lines represent baseline fairness gaps without debiasing ($\Delta DP = 0.132$, $\Delta EO = 0.118$). The final stabilised values

($\Delta DP \approx 0.046$, $\Delta EO \approx 0.038$) reflect a 65–68% reduction in bias while preserving overall recommendation performance (CTR and PQA).

The integration of the adversarial debiasing module led to substantial reductions in Demographic Parity (ΔDP) and Equality of Opportunity (ΔEO) gaps by approximately 65% and 68%, respectively, indicating that the system became significantly less sensitive to learner group attributes. This improvement suggests that the discriminator effectively suppressed latent correlations between political affiliation proxies and recommendation patterns, allowing the policy network to optimise engagement and learning outcomes in a more balanced way. Importantly, these fairness gains were achieved without compromising utility metrics. CTR and PQA both increased slightly (+1–1.5%) compared to the non-debiased baseline, demonstrating that ethical alignment can coexist with educational effectiveness when fairness constraints are embedded directly in the learning objective.

5.8 System scalability and efficiency

The HDRL system is deployed in a two-stage pipeline (ANN retrieval $\rightarrow$ RL reranking) with an average p50 latency under 50 ms and p95 under 100 ms per recommendation. To improve scalability, we apply model compression, quantisation, and knowledge distillation, enabling lightweight student models that retain over 95% of baseline accuracy while achieving 4–6 $\times$ faster inference. For large-scale or resource-limited environments, the framework supports fallback options, bandit only and distilled PPO variants that trade minimal performance for significant computational savings, ensuring practical deployment across diverse institutional infrastructures.

6 Discussion

This paper presents an HDRL framework for personalised content recommendation in IPE. The results of the HDRL performance demonstrated that it can significantly outperform the traditional systems with an 18.6% click-through rate (CTR) and 28.5% knowledge retention improvement baseline over the best performing systems. These results demonstrate the promise of reinforcement learning in the area of improving individualised learning, especially in controversial ideological areas. One of the strong points of our system is its flexibility to meet the needs of the individual learner, which presents a disadvantage to the static traditional models. The combination of DQN, PPO, and contextual bandits allows us to optimise learning paths and initial personalisation of new users. Incorporation of knowledge graphs also enhances content suggestions so that they are relevant and diverse. To add, the bias mitigation framework allows due justice to be done in the education concerning ideologies, wherein neutrality is critical. Though the results are promising, we note that HDRL has a cold-start problem, where an effective start to introducing recommendations requires not least 10 interactions. It also has a huge computational cost, and therefore, it needs to be optimised to deploy in real time. The manual validation of the quality of the knowledge graph is subjective and although our bias mitigation system is efficient, it should be changed as the ideological perception changes. Our system provides the capability of adaptive and personalised learning, with the adjustment to the needs of an individual learner, and with the neutral presentation of the political information. This methodology will promote critical thinking, a key feature of political education, and decrease the chances of ideological bias. Deploying

an HDRL-based recommender in IPE involves key ethical and practical challenges. Data privacy is ensured through anonymisation and adherence to GDPR and PIPL principles, with explicit learner consent required for future deployments. Political neutrality is maintained through adversarial debiasing and content filtering, ensuring diverse perspectives. Transparency is prioritised, allowing instructors to inspect recommendation rationales and override biased outputs. Real-world challenges include infrastructure costs, compliance with data laws, and institutional acceptance. Pilot deployments with non-graded modules and continuous fairness audits are essential for addressing these issues. The HDRL framework improves engagement and retention but has limitations. The dataset is narrow, requiring broader data for generalisation. It needs approximately 10 interactions for stable adaptation, which meta-learning could address. HDRL also has high computational costs, which can be optimised with model compression. Qualitative user feedback is needed alongside statistical tests. Addressing these will make HDRL viable for large-scale education.

7 Conclusion

This paper presented a framework for personalised IPE by combining techniques such as DQN, PPO, and Contextual Bandits, along with knowledge graphs and bias mitigation strategies. Our system successfully adapts content recommendations in real time. Experimental results demonstrate significant improvements in learner engagement, knowledge retention, and content relevance compared to traditional recommendation models. The HDRL approach proves to be an effective solution for delivering personalised, dynamic, and fair educational experiences, addressing the limitations of static systems in IPE. While challenges such as cold-start scenarios and computational demands remain, the HDRL framework offers promising directions for future advancements in personalised education systems.

Author contributions

Zhiyong Li: conceptualisation, methodology, data collection, analysis, and writing of the manuscript. Reviewed and approved the final manuscript.

Funding

The authors declare that no funds, grants, or other support were received for the conduct of this research.

Data availability

The datasets collected during the current study are available from the corresponding author on reasonable request. The data used in this study was collected from a nationwide online ideological education platform and includes interaction records and content items related to ideological and political education. For further details on data access or collaboration, please contact the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 21 August 2025 / Accepted: 2 January 2026

Published online: 23 January 2026

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